2026
Practical Parallel Computing
(実践的並列コンピューティング)

Part 1: OpenMP
No. 3: Task Parallelism

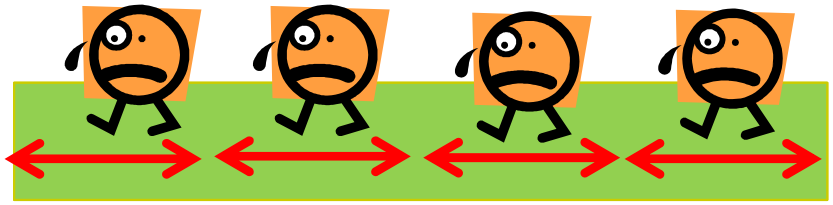
Toshio Endo
endo@scrc.iir.isct.ac.jp

Overview of This Course

- Part 0: Introduction
  - 2 classes
- Part 1: Using OpenMP (OMP) ← We are here (3/4)
  - 4 classes
  - Report (required)
- Part 2: GPU programming with OpenACC (ACC)
  - 2 classes
  - Report (required)
- Part 3: GPU programming with CUDA
  - 3 classes
  - Report (elective)
- Part 4: Using MPI
  - 3 classes
  - Report (elective)

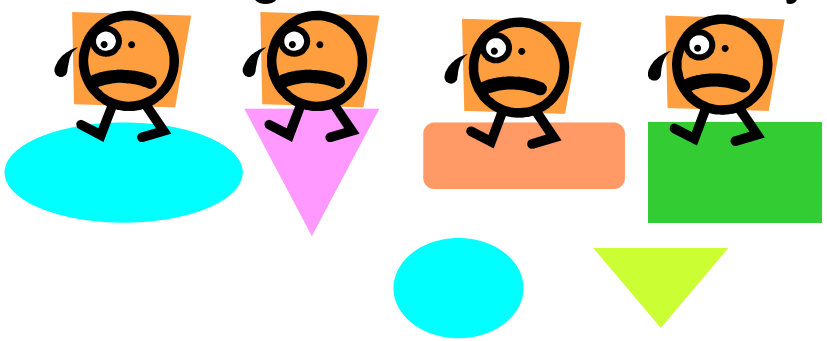
Today's Topic：Task Parallelism ~Comparison with Data Parallelism~

- Data Parallelism：
  - Every thread does uniform/similar tasks for different part of large data



cf) mm, diffusion samples

- Task Parallelism：
  - Each thread does different tasks
    - Sometimes the number of tasks is unknown beforehand
    - Sometimes tasks are generated recursively

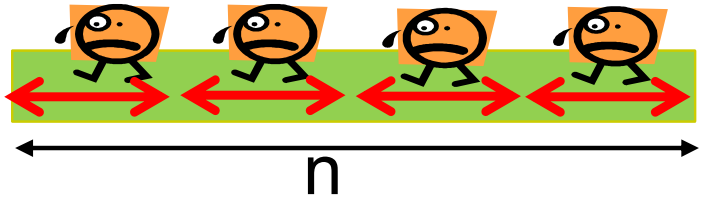


cf) fib, sort samples today

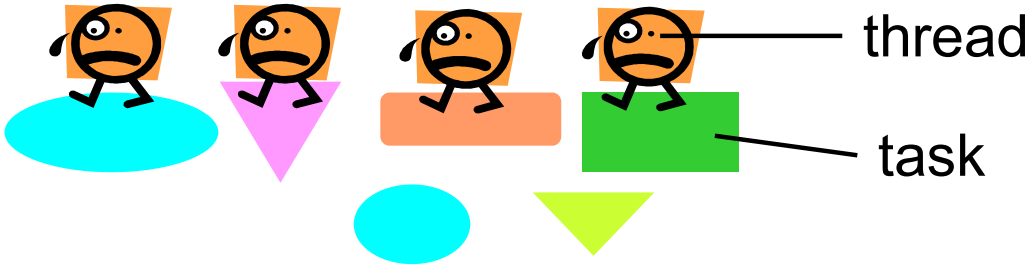
Related to assignment [O3]

Data Parallelism/Task Parallelism in OpenMP

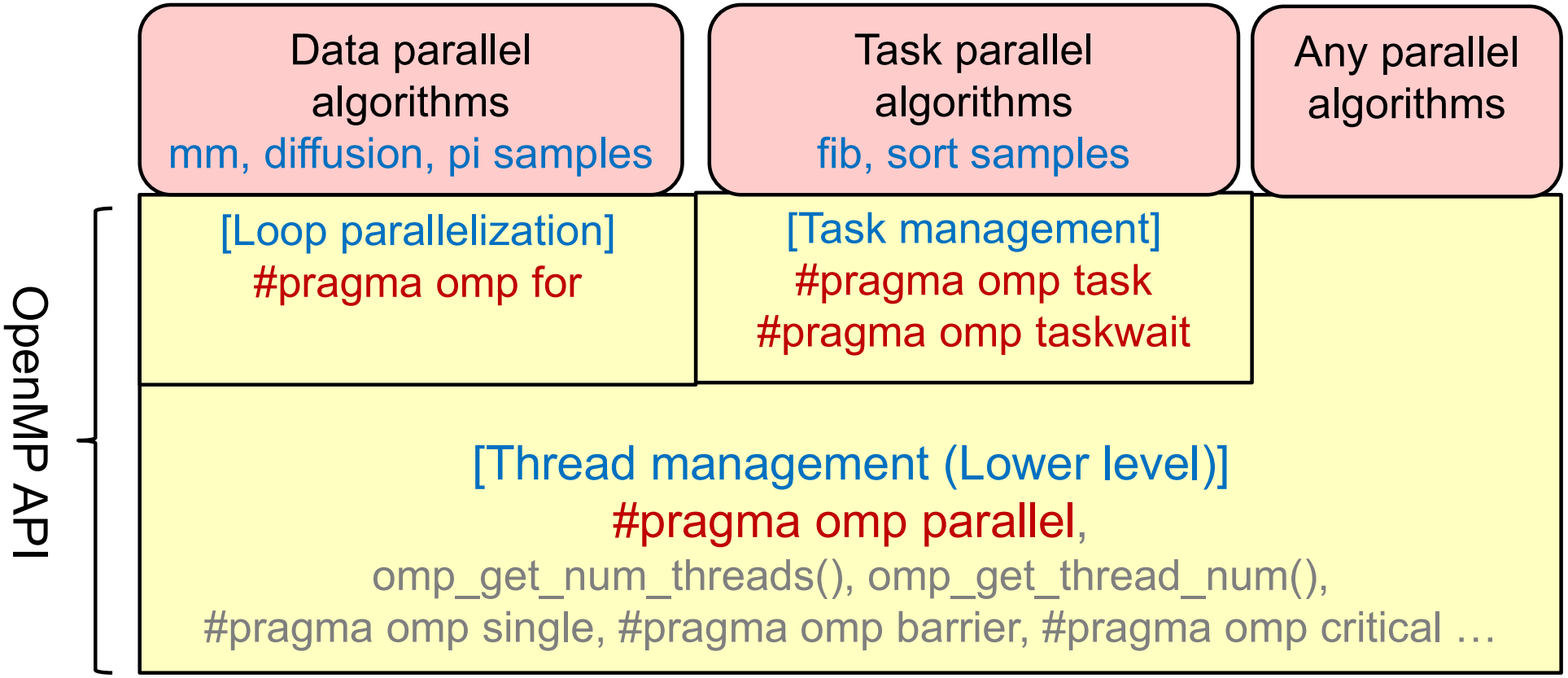
- #pragma omp for
  - Used for data parallelism (basically)
  - Number of tasks is known before starting for-loop
    - for (i = 0; i < n; i++) … → *n* tasks are divided among threads



- #pragma omp task
  - Used for task parallelism (basically)
  - Number of tasks may change during execution



Relationship of OpenMP Syntaxes



※ This grouping is different from official one
https://openmp.org/specifications/

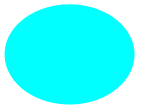
task/taskwait Syntaxes

See a sample at ppcomp-ex/omp/tasks/

```
#pragma omp parallel
#pragma omp single
{
#pragma omp task
  {
    A;
  }

#pragma omp task
  B;

  C;
#pragma omp taskwait
}
```



"task" syntax generates a task that executes the following block/sentence

- A task is executed by one of threads who is idle (has nothing to do)
- New tasks and the original task may be executed in parallel
- Recursive task generation is ok
  - A parent task generates children tasks, and one of generates grandchildren…

"taskwait" syntax waits end of all children tasks

Relations between "Tasks" and "Threads"

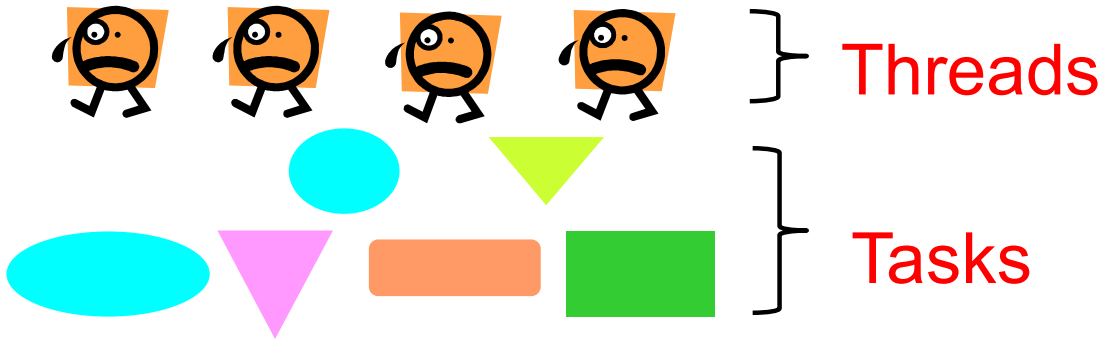
Task A and task B are executed in parallel

⟺ Confusing

Thread A and thread B are executed in parallel

Each thread executes tasks one after another

- Number of threads is (basically) constant during a parallel region
  - OMP_NUM_THREADS, usually no more than number of processor cores
- Number of tasks may be changed frequently
  - may be >>number of processor cores
- When a thread becomes idle, it takes one of tasks and executes it

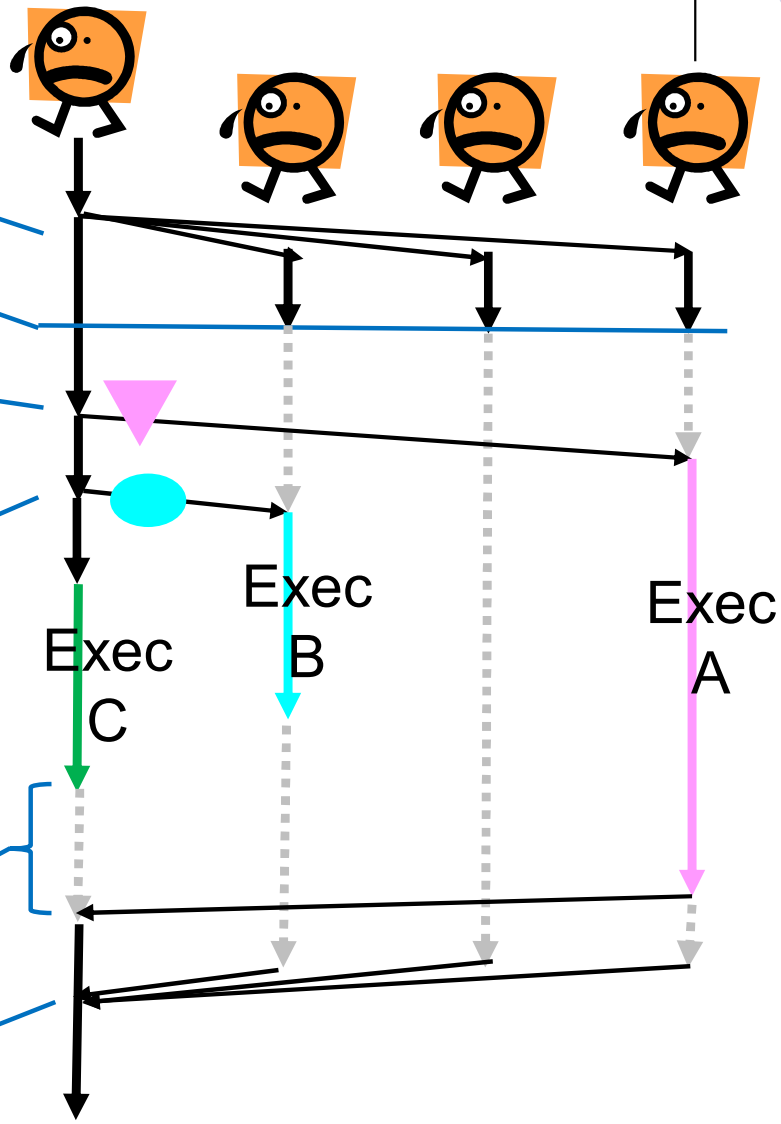


Threads Executes Tasks
(see “omp/tasks” sample)

```
#pragma omp parallel
#pragma omp single
{
#pragma omp task
  {
    A;
  }

#pragma omp task
  B;

  C;
#pragma omp taskwait
}  (end of parallel region)
```



Note on Using "task" Syntax

- In OpenMP, tasks are taken and executed by idle threads

→ We need to prepare idle threads before creating tasks

```
#pragma omp parallel        ← Multiple threads start
#pragma omp single          ← Only a single thread executes followings
  {                           (other threads become idle)

          :  (task generations)

  }                         ← Parallel region finishes
```

[Q] What if we omit "omp parallel" & "omp single"?
→ There is 1 thread, which executes all tasks
→ No speed up! ☹

[Q] What if we omit "omp single"?
→ Every thread execute all tasks redundantly
→ No speed up! ☹

About Number of Tasks

- In the tasks-omp sample, there are 3 tasks in the world
  → No speed up with ≧ 4 threads

  "Too less tasks are bad ☹"

- To use threads (CPU cores) effectively, the number of tasks should be ≧ OMP_NUM_THREADS

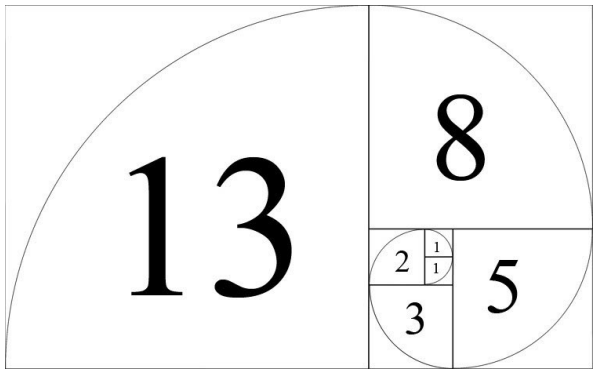
  ➔Next, we see sample programs that generates plenty of tasks

  "Too much tasks are also bad ☹"

“fib” Sample Program

sequential version

- Available at ppcomp-ex/base/fib/
  - Go to the directory and “make”
- It calculates the *Fibonacci number*
  - fib(n) = fib(n-1) + fib(n-2)
  - 1, 1, 2, 3, 5, 8, 13…
- Execution: ./fib [n]
  - ./fib 40 → outputs 40th Fibonacci number
- Recursive function call is used
  - It is an inefficient algorithm as a sample
- Computational complexity: O(fib(n)) ≒ $O(1.618^n)$
  - It is unknown before execution



Using fib Sample

- This assumes you have downloaded samples from https://github.com/toshioendo/ppcomp-ex.git
  - (details are in ppcomp26-1-2 slides)

```
cd        [If you copied sample in sub-directory, go there]
cd ppcomp-ex/base/fib
ls
[you will see 3 files of fib.c, Makefile, job.sh]
make      [this creates an executable file]
./fib 40
```

We will use omp/fib-slow and omp/fib later

Algorithm of fib

```
long fib(int n)
{
  long f1, f2;
  if (n <= 1) return n;


  f1 = fib(n-1);


  f2 = fib(n-2);



  return f1+f2;
}
```

Recursive call is used

- fib(n) depends on fib(n-1) and fib(n-2)

recursive call is stopped

recursive call

OpenMP Version of fib (version 1)

```
long fib_r(int n)
{
  long f1, f2;
  if (n <= 1) return n;

#pragma omp task shared(f1)
  f1 = fib_r(n-1);

#pragma omp task shared(f2)
  f2 = fib_r(n-2);

#pragma omp taskwait

  return f1+f2;
}
```

Tasks are generated

We wait for completion of the above 2 tasks

Available at

ppcomp-ex/omp/fib-slow

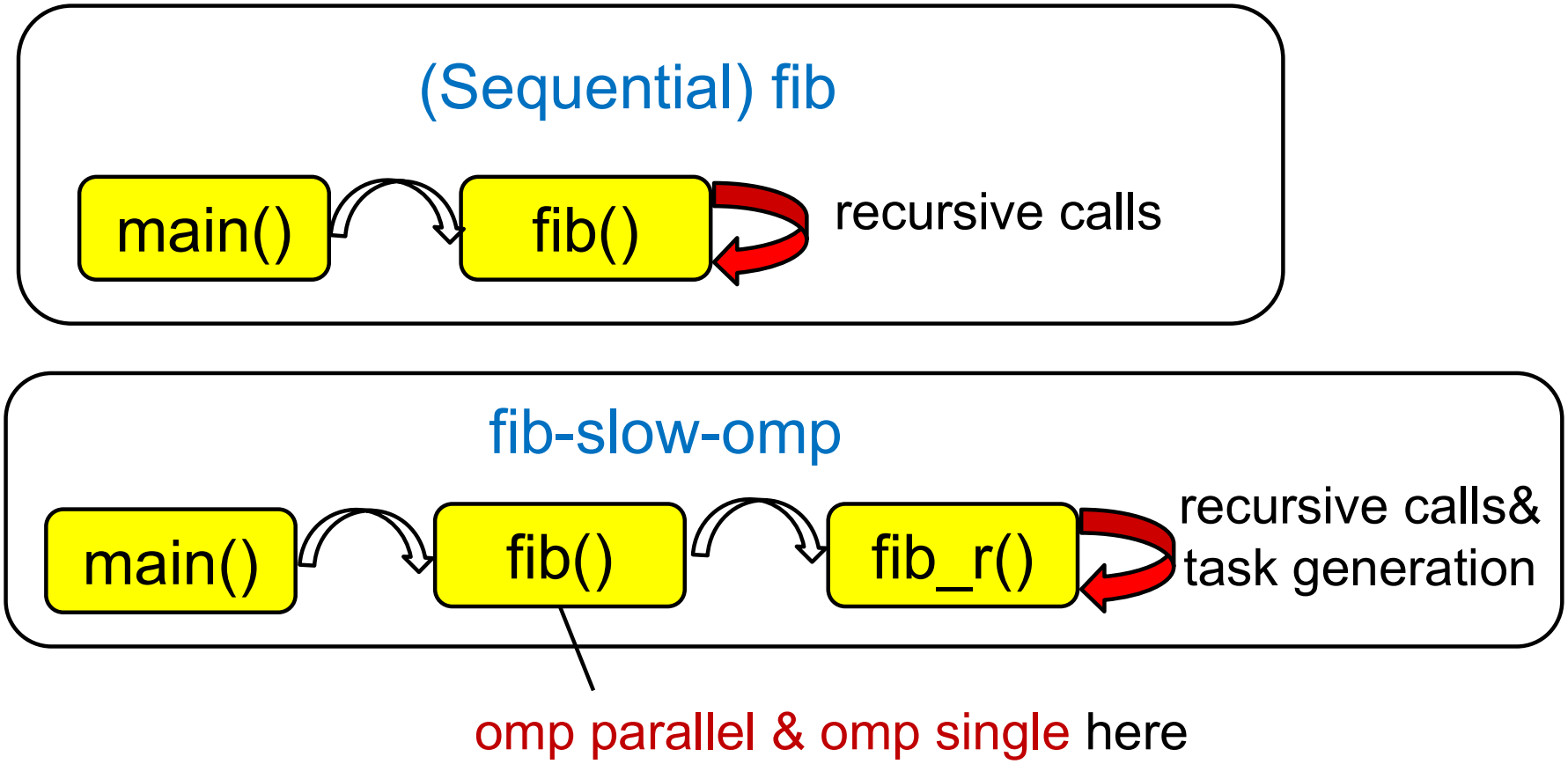
- In this version,

  a task = recursive call

Don't forget "omp taskwait"

Note on omp parallel → omp single

- We need “omp parallel & omp single” <u>only once</u>, but where?



Rules about Variables

In default, *copies* of variables are created for each child task

- The value of "*n*" is brought from parent to a child task
  → OK ☺
- But a child has a only copy → update to "f1" or "f2" is not visible to parent. NG! ☹

"shared(var)" option makes the variable "var" be shared between parent and the child

- Using it, update to "f1" or "f2" is visible to parent

The First Version is Too Slow

Execution time of ./fib 40

- On a TSUBAME4.0 interactive node (24cores)

fib

1	threads
0.60	seconds

fib-slow-omp

1	2	4	threads
13	~140	~250	seconds

- OpenMP version is much slower than original fib ☹
  - Even with 1 thread, 20x slower
- Also it is much slower with multi-threads

→ How can we improve?

Pitfall in “task” Syntax

- While OpenMP allows to generate many tasks, task generation cost is not negligible

Rough comparison:

Function call cost	<<	Task generation cost	<<	Thread generation cost

- In version 1, “./fib n” generates O(fib(n)) tasks

→ Creating too much tasks is bad!

- How can we reduce the number of tasks?

OpenMP Version of fib (version 2)

```
long fib_r(int n)
{
  long f1, f2;
  if (n <= 1) return n;

  if (n <= 30) {
     f1 = fib_r(n-1);
     f2 = fib_r(n-2);
  }
  else {
#pragma omp task shared(f1)
    f1 = fib_r(n-1);
#pragma omp task shared(f2)
    f2 = fib_r(n-2);
#pragma omp taskwait
  }
  return f1+f2;
}
```

if *n* is "sufficiently" small, we do not generate tasks

Available at

ppcomp-ex/omp/fib

- To avoid generating too many tasks, we compare n and a threshold (=30 here)
- If n is large, we generate tasks
- If n is small, we do not generate
- Changing threshold (=30) would affect performance

Performance of Version 2

Execution time of ./fib 40

fib

1	threads
0.50	seconds

fib-slow-omp

1	2	4	8	24	threads
13	~140	~250	-	-	seconds

fib-omp

1	2	4	8	24	threads
0.50	0.31	0.19	0.11	0.05	seconds

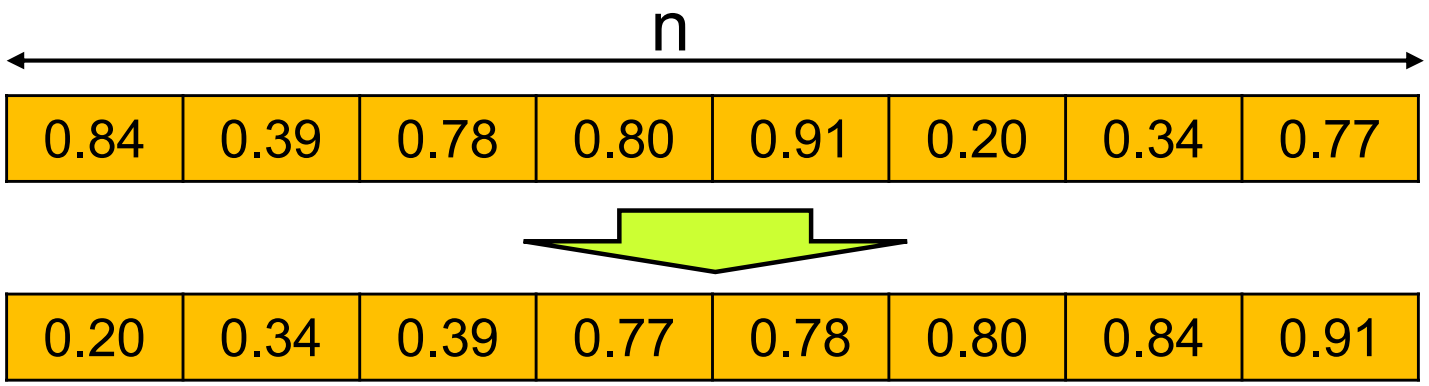
- Performance of Version 2 is largely improved and more stable

→ Restricting task generation is important for speed

“qsort” Sample Program Related to Assignment [O3]

Available at ppcomp-ex/base/qsort

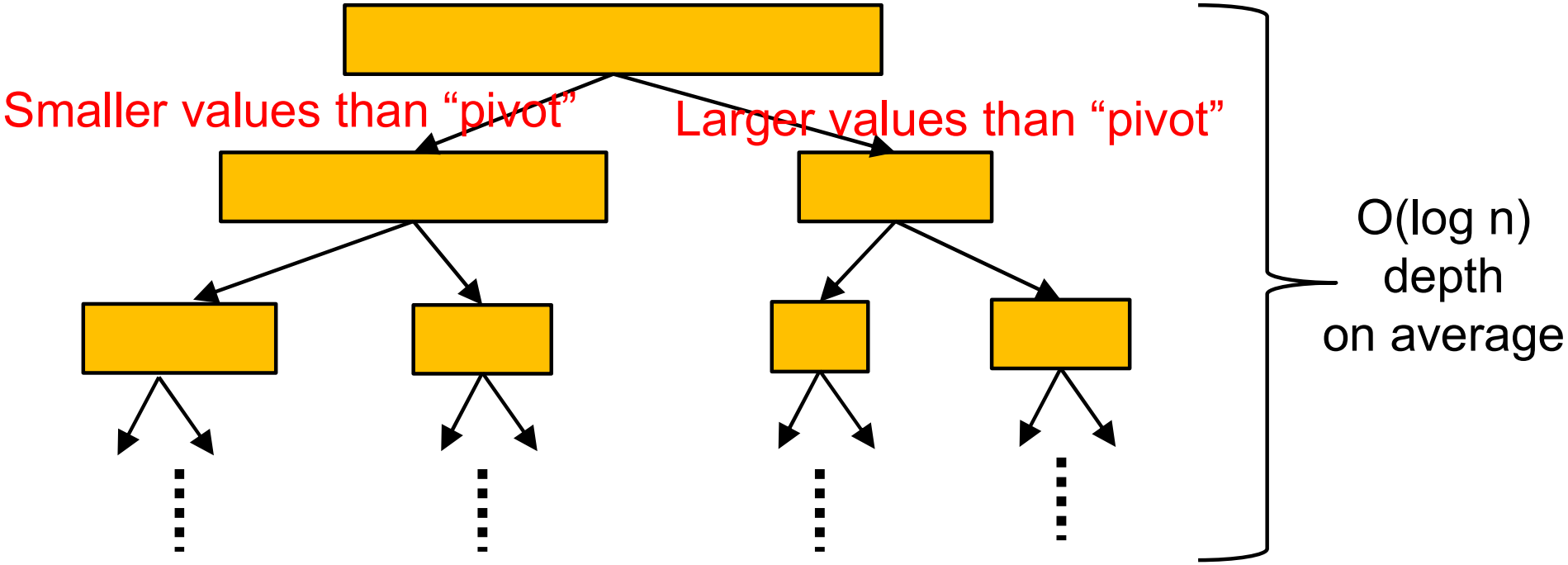
- Execution: ./qsort [n]
- It sorts an array of length *n* by the quick sort algorithm
  - Array elements have double type
- Compute Complexity: O(n log n) on average
  - More efficient than $O(n^2)$ algorithm such as bubble sort



There is also ppcomp-ex/omp/qsort
- [NOTE] qsort.c is not parallelized
- You can use it as a start point

Quick Sort

- A recursive algorithm
  - Take a value, called “pivot” from the array
  - Partition array into two parts, “small” and “large”
  - “small” part and “large” part are sorted recursively



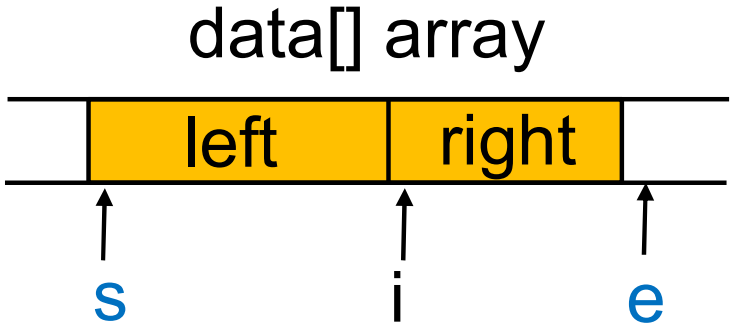
Structure of sort Sample

```
int sort(double *data, int s, int e)
{
  int i, j;
  double pivot;
  if (e-s <= 1)  return 0;

  /* pivot selection */
       :

  /* partition data[ ] into 2 parts */
       :
  /* Here "i" is boundary of 2 parts */

  sort(data, s, i);   /* Sort left part recursively*/
  sort(data, i, e);  /* Sort right part recursively */
}
```



Harder to parallelize

Generating 2 tasks would be a good idea

[Q] How can we restrict too much task generation?

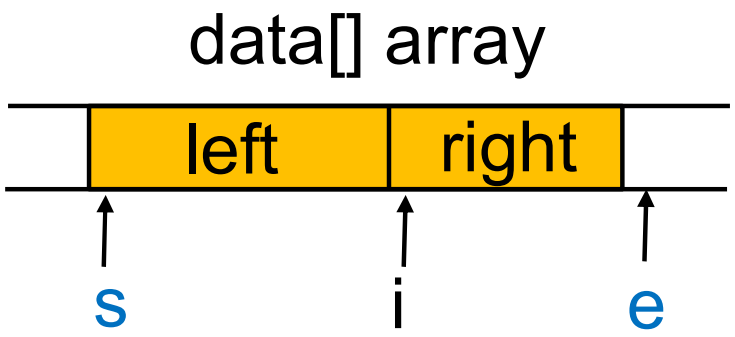
Is it Correct to Parallelize Recursive Calls in sort?

C1

C2

```
:
sort(data, s, i);  /* Sort left part recursively*/
sort(data, i, e);  /* Sort right part recursively */
```

- Let us discuss why computations C1 and C2 can be parallelized
  - Analyze read-set R and write-set W of each



- R(C1) = W(C1) = {data[s], data[s+1], … data[i-1]}
- R(C2) = W(C2) = {data[i], data[i+1], … data[e-1]}

Disjoint
➔ independent!

Even with recursive task generations, this discussion can be applied

[Revisited] When We Can Use "omp for"

- Loops with some (complex) forms cannot be supported, unfortunately ☹
- The target loop must be in the following form

```
#pragma omp for
  for (i = value; i op value; incr-part)
    body
```

```
"op" : <, >, <=, >=, etc.
"incr-part" : i++, i--, i+=c, i-=c, etc.
```

```
OK ☺:  for (x = n; x >= 0; x-=4)
NG ☹:  for (i = 0; test(i); i++)
NG ☹:  for (p = head; p != NULL; p = p->next)
```

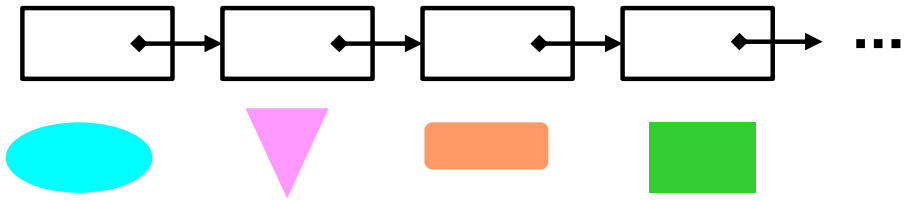
⇨ *Instead, we can parallelize it with "task" syntax*

Parallelize Irregular Loops with "task" Syntax

- In list search, number of iterations cannot be known before execution → we can use "task"

```
#pragma omp parallel
#pragma omp single
  {
     for (p = head; p != NULL;
          p = p->next) {
#pragma omp task
          [Do something with p]
     }
#pragma omp taskwait
  }
```

- A task for one list node = one OpenMP task



Note:
- The number of generated tasks = List length.

→ Consider total task generation costs

- Using TSUBAME job scheduler:
  - ppcomp-sup slides

Assignments in OpenMP Part (Abstract)

Choose one of [O1]—[O4], and submit a report

Due date: May 7 (Thu)

[O1] Parallelize “diffusion” sample program by OpenMP.

[O2] Parallelize “bsort” sample program by OpenMP.

[O3] Parallelize “qsort” sample program by OpenMP.

[O4] (Freestyle) Parallelize *any* program by OpenMP.

For more detail, please see ppcomp26-1-1 slides

Plan of Part 1 (OMP)

- No. 1
  - Introduction to OpenMP
- No. 2
  - Data parallelism with for loops
  - diffusion sample [O1], bsort sample [O2]
- No. 3
  - Task parallelism
  - qsort sample [O3]
- No. 4
  - What are bottlenecks, race conditions?